# TWO COUNTEREXAMPLES IN THE EXTREMAL THEORY OF BOOLEAN COCHAINS

PATRIZIO CINTIOLI

Abstract. This paper presents exact counterexamples to Conjectures 5.1 and 5.4 in version v5 of Jonathan Cary's *On Kleitman's Conjecture* (arXiv:2402.03150). The five-dimensional example exhibits a nonpolar global maximizer of the weighted cochain objective, with matching rational primal and dual certificates and optimal value 3. The eleven-dimensional example provides a monotone intersecting Boolean function for which no real cochain with the prescribed codifferential satisfies the directional-maxima saturation property; the obstruction has the exact positive gap 115/131072. The latter function nevertheless admits a polar cochain. These counterexamples concern two stronger structural assertions about cochains and do not contradict the recently obtained results on Chvátal's, Kleitman's, and Kahn's conjectures.

## Introduction

Chvátal's conjecture asserts that every finite downset admits a maximum-size intersecting subfamily which is a star. Kleitman's stronger conjecture concerns weighted intersecting families and equivalent flow formulations. Cary's *On Kleitman's Conjecture* [1] reformulates these questions using cochains on the discrete cube and introduces additional structural conjectures about the solutions of certain codifferential equations.

To make the terminology precise, let $n \geq 1$ and let $\mathcal{D} \subseteq 2^{[n]}$. The family $\mathcal{D}$ is a *downset* (or *hereditary family*) if $A \in \mathcal{D}$ and $B \subseteq A$ imply $B \in \mathcal{D}$. A family $\mathcal{I} \subseteq \mathcal{D}$ is *intersecting* if $A \cap B \neq \emptyset$ for all $A, B \in \mathcal{I}$. For each $i \in [n]$, the *star* of $\mathcal{D}$ centred at $i$ is $\mathcal{S}_i(\mathcal{D}) = \{A \in \mathcal{D} : i \in A\}$. In this notation, Chvátal's conjecture (now a theorem) states that

$$\max_{\substack{\mathcal{I} \subseteq \mathcal{D} \\ \mathcal{I} \text{ intersecting}}} |\mathcal{I}| = \max_{i \in [n]} |\mathcal{S}_i(\mathcal{D})|.$$

Thus *maximum-size* means largest cardinality, not merely *maximal under inclusion*: at least one full star attains the largest possible size of an intersecting subfamily of $\mathcal{D}$.

Recent preprints have substantially advanced the underlying extremal questions. Chang, Liu and Liu [2, Theorem 1.2] establish a sharp correlation inequality for *two arbitrary increasing Boolean functions*: antipodality is not assumed in the general theorem. They deduce Chvátal's conjecture and, in their discussion entitled "Kleitman's weighted bound," prove a weighted star inequality for maximal intersecting families [2, Proposition 5.3]; that discussion explicitly cites Cary. Ellis, Filmus and Friedgut [3, Theorems 1 and 2] give a short spectral proof of Chvátal's conjecture and, in their Section 3, of Kleitman's conjecture. Keevash [4, Theorem 1.1] proves Kahn's flow conjecture for increasing, antipodal, real-valued functions $h$, with the asserted flow from $L_\lambda$ to $h_+^2$.

The conclusions of these results must be distinguished from those of Cary's Conjectures 5.1 and 5.4. Correlation inequalities and weighted star or flow bounds do not assert that *every* maximizer of a given weighted cochain objective is polar, or that there *exists* a cochain simultaneously realizing the prescribed codifferential and all directional maxima on the pivotal edges. In particular, the absence of antipodality or maximality in the examples below is not, by itself, a reason to dismiss the general correlation inequality of [2]: its *conclusion* is different. The counterexamples are compatible with all three preprints.

*Date*: November 8, 2026.

2020 *Mathematics Subject Classification.* Primary 05D05; Secondary 06E30, 90C46.

*Key words and phrases.* Cary's conjectures, intersecting families, hereditary families, Boolean functions, polar cochains, primal–dual certificates, directional-maxima saturation.

More precisely, the first example gives a nonpolar *global* maximizer in five variables for positive weights, thereby falsifying the universal assertion of Conjecture 5.1. The second example, in eleven variables, excludes *every* cochain satisfying the saturation condition in Conjecture 5.4, while an explicit polar cochain still exists. Both examples are monotone and intersecting but not maximal intersecting. The precise statements of Cary's conjectures in the version studied here (*v5*, 13 September 2026) impose no maximality requirement: see pp. 21 and 22, respectively. No assertion is made here about versions with additional maximality hypotheses, about the minimal dimensions of counterexamples, or about the failure of Cary's Conjecture 4.1 or the original conjectures of Chvátal, Kleitman, and Kahn.

Section 1 collects the normalizations and two certificate lemmas. Sections 2 and 3 give the exact counterexamples; Section 4 discusses extensions and polarity, and Section 5 records reproducibility details.

## 1. Conventions and certificate principles

The arguments rest on two elementary certificate principles. One bounds a weighted cochain objective from above; the other turns a proposed saturation property into a covariance inequality.

We follow the conventions in Cary [1]. Vertices are subsets of $[n] = \{1, \dots, n\}$, with $x_i(A) = 1$ if $i \in A$ and $x_i(A) = -1$ otherwise. All vertex expectations are uniform. For real vertex functions $u, v$ we set

$$\langle u, v\rangle_0 = 2^{-n} \sum_{A\subseteq[n]} u(A)v(A), \qquad \widehat{u}(\{i\}) = \langle u, x_i\rangle_0, \qquad \mu_u = \mathbb{E}u.$$

For each $i$, a real 1-cochain has a component $\phi_i$ defined on the $2^{n-1}$ upward edges $A \to A \cup \{i\}$ with $i \notin A$. The derivative is the *semidifference*

$$(du)_i(A) = \frac{u(A \cup \{i\}) - u(A)}{2}.$$

The codifferential is

$$(\delta\phi)(A) = \sum_{i\in A} \phi_i(A \setminus \{i\}) - \sum_{i\notin A} \phi_i(A). \tag{1}$$

The scalar product on real 1-cochains uses the uniform measure *separately in each direction*:

$$\langle \phi, \psi\rangle_1 = 2^{-(n-1)} \sum_{i=1}^{n} \sum_{A\subseteq[n]\setminus\{i\}} \phi_i(A)\psi_i(A).$$

Here $\mathbb{E}_i\phi_i$ denotes the uniform mean over the $i$-edges. Regrouping the contributions of each edge at its two endpoints proves

$$\langle du, \phi\rangle_1 = \langle u, \delta\phi\rangle_0, \qquad \mathbb{E}_i(du)_i = \widehat{u}(\{i\}), \qquad \delta(dx_i) = x_i. \tag{2}$$

In particular, if $\delta\phi = f - \mu_f$, then $\mathbb{E}_i\phi_i = \widehat{f}(\{i\})$ for every $i$. A cochain is *polar* if $\phi_i(\emptyset) = \min \phi_i$ for each $i$. A function $f$ is called *polar* if it admits such a cochain with $\delta\phi = f - \mu_f$. Neither the definition nor the conjectures below impose nonnegativity or symmetry on the cochain.

**Lemma 1.1** (Weighted upper certificate)**.** *Let $\lambda$ be monotone, with $\widehat{\lambda}(\{i\}) = \alpha_i \geq 0$. For every cochain $\psi$ satisfying $\delta\psi = f - \mu_f$,*

$$\sum_i \alpha_i \min \psi_i \leq \langle \lambda, f - \mu_f\rangle.$$

*Proof.* Let $m_i = \min \psi_i$. Monotonicity gives $d\lambda \geq 0$ on all upward edges. By adjunction and the singleton identity in (2),

$$\begin{aligned}\langle \lambda, f - \mu_f\rangle_0 - \sum_i \alpha_i m_i &= \langle d\lambda, \psi\rangle_1 - \sum_i m_i\, \mathbb{E}_i(d\lambda)_i \\ &= 2^{-(n-1)} \sum_{i=1}^{n} \sum_{A\subseteq[n]\setminus\{i\}} (d\lambda)_i(A)\big(\psi_i(A) - m_i\big) \geq 0.\end{aligned}$$

This is a bound for every feasible real cochain; no compactness or strong-duality assumption is needed. □

**Lemma 1.2** (Directional-maxima obstruction)**.** *Let $f : 2^{[n]} \to \{-1, 1\}$ be monotone. Suppose $\delta\phi = f - \mu_f$ and $\phi_i(A) = M_i := \max \phi_i$ whenever $(df)_i(A) = 1$. Then every monotone real-valued $\lambda$ with $\widehat{\lambda}(\{i\}) = \widehat{f}(\{i\})$ for all $i$ satisfies*

$$\mathrm{Cov}(f, \lambda) \le \mathrm{Var}(f) = 1 - \mu_f^2.$$

*Proof.* Because $f$ is monotone and takes values in $\{-1, 1\}$, each component of $df$ takes values in $\{0, 1\}$. Saturation therefore implies

$$1 - \mu_f^2 = \langle f, f - \mu_f \rangle_0 = \langle df, \phi \rangle_1 = \sum_i M_i \widehat{f}(\{i\}).$$

Since $d\lambda \ge 0$ and $\phi_i(A) \le M_i$, with no sign condition on $\phi$ or $M_i$, we obtain

$$\mathrm{Cov}(f, \lambda) = \langle d\lambda, \phi \rangle_1 \le \sum_i M_i \widehat{\lambda}(\{i\}) = 1 - \mu_f^2.$$ □

## 2. Conjecture 5.1: a five-variable counterexample

The first counterexample combines an explicitly given nonpolar cochain with a matching monotone dual certificate. Together they establish global optimality, rather than merely feasibility.

Conjecture 5.1 of Cary [1, p. 21] asserts that, for every monotone intersecting $f : 2^{[n]} \to \{-1, 1\}$ and every strictly positive weight vector $\alpha$, *every* maximizer of $\sum_i \alpha_i \min \phi_i$ subject to $\delta\phi = f - \mu_f$ is polar. One nonpolar maximizer suffices to disprove this statement, regardless of whether other polar maximizers exist.

**Theorem 2.1.** *Let*

$$\begin{aligned} \mathcal{T} &= \{\{1,2,3\}, \{2,3,4\}, \{1,2,5\}, \{1,3,5\}, \\ &\qquad \{2,3,5\}, \{1,4,5\}, \{2,4,5\}\}, \\ \mathcal{F} &= \mathcal{T} \cup \{A \subseteq [5] : |A| \ge 4\}, \\ f &= 2\mathbf{1}_{\mathcal{F}} - 1, \qquad \alpha = (2, 1, 4, 8, 1). \end{aligned}$$

*Then $f$ is monotone and intersecting, and there is a nonpolar maximizer $\phi$ with*

$$(\min \phi_1, \min \phi_2, \min \phi_3, \min \phi_4, \min \phi_5) = \frac{1}{16}(3, 1, 4, 3, 1), \qquad \phi_5(\emptyset) = \frac{2}{16}.$$

*The optimal value of the weighted objective is* 3*. Consequently, Conjecture 5.1 of Cary is false as stated.*

*Proof.* Every member of $\mathcal{F}$ has at least three elements, so any two members intersect. Since $\mathcal{F}$ contains all four- and five-element subsets, it is upward closed. Also $|\mathcal{F}| = 7 + 5 + 1 = 13$; hence

$$\widehat{f}(\emptyset) = \frac{13 - 19}{32} = -\frac{3}{16}. \tag{3}$$

Primal certificate. Put

$$\phi_k(A) = t_k + h_k(A), \qquad (t_1, t_2, t_3, t_4, t_5) = \frac{1}{16}(3, 1, 4, 3, 1). \tag{4}$$

We specify the nonnegative $h_k$ in Table 1. For brevity, a subset $A \subseteq [5]$ is encoded as the integer $[A] = \sum_{i \in A} 2^{i-1}$. Each entry $r : s$ in the $k$th column means $16h_k(A) = s$ for $[A] = r$. All entries not listed are zero. A direct substitution into (1) gives, for every $A \subseteq [5]$,

$$(\delta\phi)(A) = \begin{cases} 19/16, & A \in \mathcal{F}, \\ -13/16, & A \notin \mathcal{F}, \end{cases} = f(A) - \widehat{f}(\emptyset). \tag{5}$$

| $k=1$ | $k=2$ | $k=3$ | $k=4$ | $k=5$ |
|---|---|---|---|---|
| 2 : 3 | 5 : 9 | 16 : 4 | — | 0 : 1 |
| 4 : 9 | 12 : 15 | 18 : 5 | — | 1 : 7 |
| 6 : 6 | 13 : 9 | 19 : 7 | — | 3 : 12 |
| 22 : 6 | 17 : 16 | — | — | 5 : 15 |
| 24 : 4 | 20 : 15 | — | — | 6 : 5 |
| 30 : 4 | 24 : 12 | — | — | 8 : 7 |
| — | 28 : 17 | — | — | 9 : 13 |
| — | 29 : 3 | — | — | 10 : 9 |
| — | — | — | — | 11 : 15 |
| — | — | — | — | 13 : 12 |

TABLE 1. All 27 nonzero coefficients $16h_k(A)$ of the primal flow.

All entries of $h$ are nonnegative and in each direction at least one entry is zero. Therefore $\min \phi_k = t_k$. But $h_5(\emptyset) = 1/16$, so $\phi_5(\emptyset) = 2/16 > \min \phi_5 = 1/16$: the cochain is not polar. Its objective equals

$$\sum_{k=1}^{5} \alpha_k t_k = \frac{2\cdot 3 + 1\cdot 1 + 4\cdot 4 + 8\cdot 3 + 1\cdot 1}{16} = 3. \tag{6}$$

Dual certificate. Define the increasing families

$$\mathcal{U}_0 = \{A : 4 \in A\}, \qquad \mathcal{U}_1 = \{A : \{3,4\} \subseteq A\},$$
$$\mathcal{U}_2 = \{A : \{1,2,4\} \subseteq A \ \vee \ \{1,3,4\} \subseteq A \ \vee \ \{3,4,5\} \subseteq A\}$$

and put

$$\lambda = 4\mathbf{1}_{\mathcal{U}_0} + 8\mathbf{1}_{\mathcal{U}_1} + 16\mathbf{1}_{\mathcal{U}_2}. \tag{7}$$

Thus $\lambda$ is monotone. Its singleton Fourier coefficients are readily computed from the three rows

$$(\widehat{\mathbf{1}_{\mathcal{U}_0}}(\{k\}))_{k=1}^5 = (0,0,0,\tfrac{1}{2},0),$$
$$(\widehat{\mathbf{1}_{\mathcal{U}_1}}(\{k\}))_{k=1}^5 = (0,0,\tfrac{1}{4},\tfrac{1}{4},0),$$
$$(\widehat{\mathbf{1}_{\mathcal{U}_2}}(\{k\}))_{k=1}^5 = (\tfrac{1}{8},\tfrac{1}{16},\tfrac{1}{8},\tfrac{1}{4},\tfrac{1}{16}).$$

It follows that

$$(\widehat{\lambda}(\{k\}))_{k=1}^5 = (2,1,4,8,1) = \alpha. \tag{8}$$

Moreover, using $\widehat{f}(\emptyset) = -3/16$, counting the vertices of each $\mathcal{U}_i$ in $\mathcal{F}$ and its complement yields

$$\langle \mathbf{1}_{\mathcal{U}_0}, f - \widehat{f}(\emptyset)\rangle = \frac{3}{32},$$
$$\langle \mathbf{1}_{\mathcal{U}_1}, f - \widehat{f}(\emptyset)\rangle = \frac{7}{64},$$
$$\langle \mathbf{1}_{\mathcal{U}_2}, f - \widehat{f}(\emptyset)\rangle = \frac{7}{64}.$$

Consequently,

$$\langle \lambda, f - \widehat{f}(\emptyset)\rangle = 4\frac{3}{32} + 8\frac{7}{64} + 16\frac{7}{64} = 3. \tag{9}$$

By Lemma 1.1, or directly for any feasible $\psi$,

$$\langle \lambda, \delta\psi\rangle = \langle d\lambda, \psi\rangle \geq \sum_{k=1}^{5} \widehat{\lambda}(\{k\}) \min \psi_k = \sum_{k=1}^{5} \alpha_k \min \psi_k,$$

since $d\lambda \geq 0$ (monotonicity). Equation (9) therefore bounds the primal objective by 3. Together with (6), this proves global optimality and the claimed counterexample. □

## 3. Conjecture 5.4: an eleven-variable counterexample

The second counterexample is a nonexistence result. An exact covariance obstruction rules out every saturating cochain; a separate argument with directional maxima gives the contradiction directly.

Conjecture 5.4 of Cary [1, p. 22] asserts that every monotone intersecting $f : 2^{[n]} \to \{-1, 1\}$ admits *at least one* real 1-cochain $\phi$ with $\delta\phi = f - \mu_f$ such that, for every $i$ and every $A \subseteq [n]\setminus\{i\}$, one has

$$(df)_i(A) = 1 \quad \Longrightarrow \quad \phi_i(A) = \max_{B\subseteq[n]\setminus\{i\}} \phi_i(B).$$

The maxima are direction-dependent; no symmetry restriction is imposed. The conjecture does not require maximality of the intersecting family. To refute it one must exclude *all* cochains satisfying both requirements, not merely exhibit a nonsaturating cochain. Let $B_1 = \{1, 2, 3, 4, 5\}$, $B_2 = \{6, 7, 8, 9, 10\}$, and $z = 11$. On $2^{[10]}$, define

$$\mathcal{H} = \{S \subseteq [10] : B_1 \subseteq S \text{ or } B_2 \subseteq S \text{ or } |S| \geq 8\}.$$

On $2^{[11]}$, put

$$\mathcal{F} = \{S \cup \{z\} : S \in \mathcal{H}\}, \qquad f = 2\mathbf{1}_{\mathcal{F}} - 1, \qquad \mathcal{G}_j = \{A \subseteq [11] : B_j \cup \{z\} \subseteq A\}.$$

Both conditions defining $\mathcal{F}$ are increasing, so $f$ is monotone. All members of $\mathcal{F}$ contain $z$, so $\mathcal{F}$ is intersecting. Also $\mathcal{G}_1, \mathcal{G}_2 \subseteq \mathcal{F}$.

**Theorem 3.1.** *The function $f$ above does not admit any real 1-cochain satisfying the divergence and directional-maxima conclusion of Conjecture 5.4 in* [1].

*Proof.* The number of members of $\mathcal{H}$ of each size is as follows:

| $\lvert S\rvert$ | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|
| $\lvert\{S \in \mathcal{H} : \lvert S\rvert = j\}\rvert$ | 2 | 10 | 20 | 45 | 10 | 1. |

There are no members of size below 5. Consequently

$$|\mathcal{H}| = |\mathcal{F}| = 88, \qquad \mu = \frac{2 \cdot 88}{2048} - 1 = -\frac{117}{128}, \qquad \mathrm{Var}(f) = \frac{2695}{16384}. \tag{10}$$

Permutations within the two blocks and interchange of the blocks act transitively on [10] and preserve $\mathcal{H}$. Furthermore,

$$\sum_{S\in\mathcal{H}} (2|S| - 10) = 2 \cdot 10 + 4 \cdot 20 + 6 \cdot 45 + 8 \cdot 10 + 10 = 460.$$

Thus $\sum_{S\in\mathcal{H}} x_i(S) = 46$ for every $i \leq 10$, and

$$\widehat{f}(\{i\}) = \frac{2 \cdot 46}{2048} = \frac{23}{512} \quad (i \leq 10), \qquad \widehat{f}(\{z\}) = \frac{2 \cdot 88}{2048} = \frac{11}{128}. \tag{11}$$

Define the real-valued function

$$\lambda = \frac{11}{4}(\mathbf{1}_{\mathcal{G}_1} + \mathbf{1}_{\mathcal{G}_2}) + \frac{1}{512}\sum_{i=1}^{10} x_i. \tag{12}$$

It is monotone, being a positive linear combination of monotone functions. The indicator of a principal upset requiring six coordinates has mean $1/64$ and singleton Fourier coefficient $1/64$ on each of those coordinates, and 0 on every other coordinate. Hence

$$\widehat{\lambda}(\{i\}) = \frac{11}{4}\frac{1}{64} + \frac{1}{512} = \frac{23}{512} \quad (i \leq 10), \qquad \widehat{\lambda}(\{z\}) = 2\frac{11}{4}\frac{1}{64} = \frac{11}{128}.$$

These agree with (11). Because $\mathcal{G}_j \subseteq \mathcal{F}$,

$$\langle \mathbf{1}_{\mathcal{G}_j}, f - \mu\rangle = \frac{1 - \mu}{64} = \frac{245}{8192}.$$

Using (12), we conclude that

$$\begin{aligned}\mathrm{Cov}(f,\lambda) &= \frac{11}{4}\cdot 2\cdot\frac{245}{8192}+\frac{1}{512}\cdot 10\cdot\frac{23}{512}\\ &= \frac{2695}{16384}+\frac{115}{131072} = \mathrm{Var}(f)+\frac{115}{131072} > \mathrm{Var}(f).\end{aligned}$$

This contradicts Lemma 1.2 and proves the theorem. □

3.1. **A direct contradiction with directional maxima.** The obstruction also has a direct formulation. Suppose $\phi$ satisfied the conjecture, put $M_i = \max \phi_i$, and let $T = \sum_{i=1}^{10} M_i$. Taking the mean in direction $i$ gives

$$M_i \geq \mathbb{E}\phi_i = \langle dx_i, \phi\rangle = \langle x_i, \delta\phi\rangle = \widehat{f}(\{i\}).$$

Thus $T \geq 10\cdot 23/512 = 115/256 > 0$. On the other hand, saturation on $df$ and (11) imply

$$\frac{23}{512}T+\frac{11}{128}M_z = \frac{2695}{16384}, \qquad \text{or} \qquad \frac{23}{44}T+M_z = \frac{245}{128}.$$

Testing against $\mathbf{1}_{\mathcal{G}_j}$, whose derivative is nonnegative, yields

$$\frac{245}{8192} = \langle d\mathbf{1}_{\mathcal{G}_j}, \phi\rangle \leq \frac{1}{64}\left(M_z+\sum_{i\in B_j} M_i\right).$$

Averaging the two inequalities gives $M_z + T/2 \geq 245/128$. Subtracting the preceding saturation equality yields $-T/44 \geq 0$, so $T \leq 0$, a contradiction. This argument does not assume that the hypothetical cochain is symmetric.

## 4. Scope, extensions, and separation from polarity

Both counterexamples extend to arbitrarily many additional coordinates. The eleven-variable example also separates the existence of a polar cochain from saturation on every pivotal edge.

**Corollary 4.1.** *The assertion of Conjecture 5.1 fails in every dimension $n \geq 5$, and that of Conjecture 5.4 fails in every dimension $n \geq 11$.*

*Proof.* For the first example, extend $f$ and the first five components of $\phi$ independently of the extra variables, and set all new components to zero. Extend the dual function by $\lambda_n = \lambda + \sum_{i=6}^{n} \varepsilon_i x_i$ with $\varepsilon_i > 0$. Its new singleton coefficients are positive and its extra terms are orthogonal to $f - \mu_f$. Both certificates retain value 3, and the fifth component remains nonpolar. For the second example, extend $f$ and $\lambda$ independently of all new variables. Their new singleton coefficients are zero and the same strict covariance gap remains. Lemma 1.2 still applies. In either construction, intersection of the old winning sets ensures intersection after adjoining dummy variables. □

Neither family is maximal intersecting. In the first example one can add $\{1,2,4\}$ while preserving the intersecting property; in the second, the family is contained strictly in the full star centered at $z$. Both conjectures, as formulated in v5, allow nonmaximal families. These examples do not decide variants restricted to maximal intersecting families, and we make no claim that dimensions 5 and 11 are minimal. The failure of the directional-maxima saturation property does not preclude the existence of a polar cochain. The following proposition makes this distinction explicit for the eleven-variable counterexample.

**Proposition 4.2.** *The function used to disprove Conjecture 5.4 is polar.*

*Proof.* Write $q = \mathbf{1}_{\mathcal{H}}$ on $2^{[10]}$ and $a = \mathbb{E}q = 11/128$. Fix the coordinate order $1,\dots,10$ and set

$$Q_i(x_1,\dots,x_i) = \mathbb{E}[q \mid x_1,\dots,x_i], \qquad Q_0 = a, \qquad Q_{10} = q.$$

For a fixed prefix $x_{<i} = (x_1,\dots,x_{i-1})$, define

$$Q_i^{\pm}(x_{<i}) = \mathbb{E}_{x_{>i}} q(x_{<i}, \pm 1, x_{>i}), \qquad \psi_i(x_{<i}) = \frac{Q_i^{+}(x_{<i}) - Q_i^{-}(x_{<i})}{2}.$$

Equivalently, $\psi_i$ is the expectation over future coordinates of the semidifference $\partial_i q = (q|_{x_i=1} - q|_{x_i=-1})/2$. Monotonicity of $q$ gives $\psi_i \geq 0$. At a fixed prefix, $Q_{i-1} = (Q_i^+ + Q_i^-)/2$, whereas $Q_i = Q_i^+$ if $x_i = 1$ and $Q_i = Q_i^-$ if $x_i = -1$. Hence, *pointwise*,

$$Q_i - Q_{i-1} = x_i\psi_i, \qquad \sum_{i=1}^{10} x_i\psi_i = Q_{10} - Q_0 = q - a.$$

Interpret $\psi_i$ as a cochain component independent of $x_i$ and of the later coordinates. Formula (1) then gives $\delta\psi = \sum_i x_i\psi_i = q - a$, not merely an identity in mean.

On the eleven-dimensional cube, $f = (1 + x_z)q - 1$ and $\mu_f = a - 1$. Extend $\psi$ independently of $x_z$ and define

$$\Phi_i = (1 + x_z)\psi_i \quad (i \leq 10), \qquad \Phi_z = a.$$

These are legitimate cochain components: $x_z$ is constant along any edge in direction $i \leq 10$. By (1),

$$\delta\Phi = (1 + x_z)(q - a) + ax_z = (1 + x_z)q - a = f - \mu_f.$$

Every old component is nonnegative and vanishes on the face $x_z = -1$, in particular at the edge adjacent to $\emptyset$; the last component is constant. Thus $\Phi$ is polar. The proof of Theorem 3.1 shows separately that this cochain cannot satisfy the stronger saturation condition of Conjecture 5.4. □

The first counterexample concerns *all weighted maximizers*, not existence of a polar cochain. The second explicitly separates polarity from the directional-maxima property. In particular, neither argument refutes [1, Conjecture 4.1], Kleitman's conjecture, or Chvátal's conjecture.

## 5. Reproducibility and scope of finite checks

The following exact-arithmetic Python programs are provided with the preprint as supplementary files in the `anc/` directory:

`verificatore_indipendente_Cary.py` and `verificatore_indipendente.py`.

Both are provided unchanged from the two earlier computational audits. They use only the Python standard library, integer arithmetic and `fractions.Fraction`, and neither uses numerical optimization. The first program (Python 3.10 or later) may be run by

```
python3 verificatore_indipendente_Cary.py --output-dir results
```

It independently reconstructs both set families, checks the primal table and its divergence on every vertex, the component minima, the weighted dual certificate, the singleton coefficients and strict correlation gap, and the conditional-expectation cochain. It also checks finite instances of dummy-coordinate extensions and of the direct maxima contradiction. The second program supplies further checks of the Fourier spectra and several coordinate orders for the polar construction. Both have checks that remain active when executed with `python3 -O`. The output files can be regenerated; see the accompanying `anc/REPRODUCIBILITY.md` for commands and file hashes.

The two programs contain the 27 entries of the primal table as input data and verify the resulting finite identities; they do not parse the LaTeX manuscript. All computations concern finite certificates. The lemmas in Section 1, the logical quantifiers in Cary's conjectures, and the extensions to every higher dimension are justified by the mathematical arguments in the text, not by finite enumeration. The computational audits do not constitute formal proof-assistant verification or external journal peer review.

## Acknowledgments

The two counterexamples were discovered by ChatGPT 6 Pro (OpenAI), which also assisted in developing their proofs and preparing the manuscript. The results were subsequently examined through separate AI-assisted mathematical audits using ChatGPT 6 Pro and GPT-6 Astra, followed by a cross-audit. The author takes full responsibility for the mathematical statements and the final content of the paper.

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Mathematics Division, School of Science and Technology, University of Camerino, 62032 Camerino, Italy

*Email address*: patrizio.cintioli@unicam.it

Personal webpage: https://patrizio-cintioli.github.io/